ZAREEN WARRAICH…..FUTURE ZOOLOGIST GROUP.

ZOO502 - Animal Physiology and Behavior

Current paper solved question final term February 2020

1. How alveolar collapse is prevented? 2 marks

Prevention of Alveolar Collapse:

Alveolar collapse normally does not occur for two reasons

- Surrounding tissue prevents over-expansion of alveoli.
- Presence of pulmonary surfactants.

2. Peristalsis 2 marks

The esophagus conducts the bolus by peristaltic movement.

Peristaltic movements are the waves of contraction and relaxation in the smooth muscles of alimentary canal walls. These movements start from oral cavity and end at the distal end of rectum.

3. Three use of crops 3 marks

In some animals, esophagus contains a sac-like expanded section, the crop, which is used to store food before digestion.

Crop is generally found in those animals that feed infrequently. It allows quantities of food to be stored for digestion at a later time

Many birds e.g. chicken.

Leeches, which feed very infrequently, with weeks or months between feeding periods.

In some animals crops are also used to ferment or digest foods for purposes other than normal digestive process. For example some parent birds prepare food in this way to be regurgitated for their nestlings e.g. pigeon's milk.

4. Write three type of blood vessels 3 marks

network of blood vessels

1) arterioles,
2) capillaries
3) venules

and a network of lymph vessels, the largest of which is the central lacteal.

Blood Vessels

Three types of blood vessels carry blood during circulation:

- • Arteries which carry blood away from the heart to organs and tissues.
- • Veins which return blood to the heart.
- • Capillaries which have thin walls formed of only a single layer of endothelial cells and permit exchange of materials between blood and tissues.

5. Mechanism of panting 3 marks

Temperature zone which activates evaporative heat-dissipating mechanisms.

Animals react to higher temperature by losing heat by active heat-dissipating mechanisms such as sweating or panting

temperatures above the upper critical temperature lead to hyperthermia. Hyperthermia occurs as the heat production and gain exceeds the rate of heat loss.

If evaporative heat loss mechanisms fail, hypothermia becomes lethal

Panting

- Panting involves use of respiratory system to lose heat by evaporative cooling. It occurs in mammals, birds and some reptiles.
- During panting, mammals inhale through the nose and exhale through the mouth, exposing the tongue and other mouth structures to encourage further water evaporation and therefore heat loss.
- It also involves increase in breathing rate and depth (hyperventilation).
- Heat is carried away in exhalant air because the dimensions of the mouth are such that exhalant air retains the heat absorbed in the lungs.
- In contrast, nasal passages and their vascularization are effective in retaining both water and body heat

6. Role of pheromones 10 marks

Pheromones:
Chemical messengers released by many animals in environment to communicate. Spread through air.
Characteristics:

a. Affect the behavior of individuals of a species
b. Effective at remarkably low concentrations
c. Have particular odors
d. Detected by olfactory receptors of receiving animal **Roles of Pheromones:**

1. Recognize Members of Species:
Unicellular organisms recognize members of their own species. Many insects identify members of their colony.

2. Reproductive Roles:
Pheromones attract members of opposite sex for breeding.
Induce courtship behaviors in members of a species.
Induce spawning behavior in aquatic animals with external fertilization.

3. Territorial Behaviors:
Many male mammals mark their territories with pheromones. It warns
other males of their occupied area. **4. Insect's Social Order:**
In a honeybee colony, pheromones produced by queen maintain hive's complex social order.

5. Alarm Calls:
Members of some species produce pheromones on getting injured.
It warns other members by the presence of danger in area.

6. Repel Predators:

Pheromones produce by the pray may repel predators.
Example: The skunks produce foul-smelling musk that makes skunks unpalatable or distasteful to their enemies.

7. What is kin selection give example 5 marks

Kin Selection

- Kin selection is an altruistic evolutionary strategy that favors the reproductive success of an organism's close relatives, even at a cost of organism's own survival and reproduction.
- Kin selection is an instance of inclusive fitness, which combines the number of offspring produced by an individual itself and the number of individuals it can ensure by supporting other close relatives (siblings).
- William Hamilton proposed the idea of kin (relatives) selection to explain how selection acting on related animals can affect the fitness of an individual.
- Through kin selection, a gene carried by a particular individual passes to the next generation through a related individual.
- The fitness of an individual is based on the genes it passes on as well as on those common genes that its relatives pass on.
- For kin selection to be effective, the individuals of a group must be able to identify their relatives. This is the reason that it is found in animals with advanced social structures i.e. primates and social insects.
- The altruistic kin selection behavior is most readily apparent in the act of parents sacrificing for their offspring. When parents sacrifice their own well-being to produce and aid offspring, this actually increases the fitness of the parents because it maximizes their genetic representation in the population.

8. four respiratory pigments 2 marks

A **respiratory pigment** is a molecule, such as hemoglobin in humans and other vertebrates, that increases the oxygen-carrying capacity of the blood. The four most common invertebrate **respiratory pigments** are

i. hemoglobin,
ii. hemocyanin, iii. hemerythrin iv. chlorocruorin.

9. 4 accessory glands of alimentary canal 3 marks

Accessory Glands: Secrete digestive juices through ducts.
In mammals, the alimentary canal is associated with various accessory glands that secrete digestive juices through ducts into the gut. The accessory glands of the mammalian digestive system are:

a. Three pairs of salivary glands
b. Gastric glands
c. Pancreas
d. Liver
e. Gallbladder

10. Aggregation with example 2 marks

Aggregation

- An aggregation is any form of gathering of animals or the process of coming together.
- There is minimal interaction between members of an aggregation.

For example, several Drosophila flies living on a piece of rotting fruit

11. Difference between cortex and medulla 3 marks

Cortex and medulla

The kidney is divided into two functional layers that contain the functional urine forming units— nephrons. The outer layer is called renal cortex while the inner layer is renal medulla.

12. Torpor with example 5 marks

Torpor

Small endotherms can not maintain their high rates of metabolic rates when they are not feeding during periods of inactivity.

During those periods, some of these animals enter a state of torpor, in which body temperature and metabolic rate decrease.

The lower basal metabolism lowers the rate of conversion of energy stores into body heat.

Thus, it is generally advantageous to allow body temperature to decrease during periods of nonfeeding and inactivity.

As the animal comes out of torpor and becomes active, it undergoes a burst of metabolic activity, through shivering or oxidation of brown-fat and its body temperature rises.

Daily torpor is practiced by many terrestrial birds. The hummingbird is a classic example, allowing its body temperature to fall from a daytime level of about 40°C to a nighttime level as low as 13°C.

Several species of small mammals also undergo torpor (e.g., shrews).

13. Role of Appendix in digestion and function 3 marks

The appendix is a finger-like, blind-ended tube connected to the cecum, from which it develops in the embryo. The cecum is a pouch-like structure of the colon, located at the junction of the small and the large intestines.

Appendix has no role in digestion or absorption but has an undefined role in immunity as it has many lymph nodes

14. Types of stomach 5 marks

Types of Stomach in Vertebrates

Based on the number of chambers, stomachs are classified as:

- Monogastric stomachs
- Digastric stomachs **Monogastric Stomachs**

A Monogastric stomach consists of a single strong muscular tube or sac.

Carnivorous and omnivorous vertebrates have a Monogastric stomach.

Digastric Stomachs

Digastric stomachs are multi-chambered and are found in ruminant mammals of order Artiodactyls e.g. deer, giraffe, sheep, cattle, camel, llama etc.

These stomachs have four chambers which are separated into two divisions.

- The first division consists of fermentation chambers called rumen and reticulum.

The second division comprises the omasum and abomasum (true stomach).

15. 4 steps of gas transport system 5 marks

A gas-transfer system in animals takes in oxygen from the environment takes it to the issues through blood and brings CO2 from tissues, transporting it to the gas exchange surface and removes it by diffusion.

Such a system involves four basic steps:

1. Breathing movements, which assure a continual supply of air or water to the respiratory surface (e.g., lungs or gills)
2. Diffusion of O2 and CO2 across the respiratory epithelium
3. Bulk transport of gases by the blood
4. Diffusion of O2 and CO2 across capillary walls between blood and mitochondria in tissue cells

16. Apoenzyme and function

Apoenzymes- An enzyme that requires a cofactor but does not have one bound. An apoenzyme is an inactive enzyme, activation of the enzyme occurs upon binding of an organic or inorganic cofactor. The term holoenzyme is used to refer to the apoenzyme bound to a cofactor.

Or

Apoenzymes refers to inactive form of enzyme which activates upon the binding of a cofactor.

Function;

It is responsible for reaction

17. Define cognition map 3 marks

Cognitive Maps is also called mental map

Some animals guide their activity by a cognitive map, a representation in the nervous system of the spatial relationships between objects in an animal's surroundings. Rather than relying solely on moving from landmark to landmark, animals using cognitive maps can navigate more flexibly and efficiently by relating landmark positions to one another.

It is essential in environment wehere strategic decisions are made in rapidly shifting economic environment

18. Neural plasticity and explain it with example 10 marks

Although the basic architecture of the nervous system is established during embryonic development, the behavioral patterns can modify with experience. This capacity of nervous system to change with experience is known as neuronal plasticity.

Neuronal plasticity is of premier importance for the survival of any organism.

Examples of Neural Plasticity

Learning, memory and development of motor skills and habits.

- Neural plasticity lies behind human intelligence.
- The ability of all higher animals to respond to stimuli in diverse ways, different from fixed and programmed reflexes.
- Behavioral plasticity, virtually demonstrated by all animals.

Mechanism of Neuronal Plasticity

Much of the reshaping of the nervous system occurs at synapses.

The mechanisms that underlie synaptic plasticity are currently the subject of many experiments.

Modifying Synapses—Synaptic Plasticity

Synaptic plasticity may takes place as a result of developmental events over the course of a lifetime. Synaptic connections that are established in embryos are later refined into adult patterns.

19. Objectives of ethology 3 marks

Ethology: "Scientific study of animal behavior in natural environment

Objectives of Ethology: Involves understanding of

1. Understanding the stimuli that elicit behavior.
2. Understanding the physiological mechanisms that mediate the response.
3. Understanding how an animal's experience during growth and development influence the response.
4. Understanding how the behavior aids in survival and reproduction.
5. Understanding the behavior's evolutionary history.

20. What is Blubber and its role 2 marks

Blubber is a thick layer of fat, also called adipose tissue, directly under the skin of all marine Mammals.

Role: 1. Blubber is a good insulator because it has a lower thermal conductivity.

2. Fatty tissues are metabolically very inactive and have little blood vascularization. This 3333prevents heat loss through body surface.
3. It stores energy, insulates heat and buoyancy.

21. Fahreous Lindquist effect 2 marks

Fahraeus-Lindqvist Effect:

Phenomenon of plasma skimming & resultant increased blood flow velocity is called Fahreous Lindquist effect. It reduces energy required to drive blood through microcirculation.

22. Hepatocytes nutrients transport 3 marks

Transport of Nutrients

Digestion products absorbed enter either the blood or the lymphatic circulation in the villi. **Blood**

Transport of Sugars and Amino Acids

- Sugars and amino acids primarily enter the capillaries of the villi.
- These capillaries are drained by venules that lead into the hepatic portal vein.
- The hepatic portal vein takes the blood from the intestine directly to the liver.
- In the liver, much of the glucose is taken up by hepatocytes and, under the influence of insulin, is converted into glycogen granules for storage. Only a controlled quantity is released into the circulation to be utilized by tissues.

23. Write note on types of stomach vertebrates 5 marks

Types of Stomach in Vertebrates

Based on the number of chambers, stomachs are classified as:

- Monogastric stomachs
- Digastric stomachs

Monogastric Stomachs

A monogastric stomach consists of a single strong muscular tube or sac. Carnivorous and omnivorous vertebrates have a monogastric stomach.

Digastric Stomachs

Digastric stomachs are multichambered and are found in ruminant mammals of order Artiodactyla e.g. deer, giraffe, sheep, cattle, camel, llama etc.

These stomachs have four chambers which are separated into two divisions.

- The first division consists of fermentation chambers called rumen and reticulum.
- The second division comprises the omasum and abomasum (true stomach).

24. Hypoxia in an aquatic environment 5 marks

Hypoxia

- oxygen deficiency in biotic envirоonment“ oxygen depletin in aqutic environment as dissolved oxygen.

Hypoxia in Aquatic Environment

- O2 levels in aquatic environment vary more frequently
- Aquatic animals are faced with frequent and rapid changes in oxygen levels. Many aquatic Animals can withstand very long periods of hypoxia

Respiratory Responses of Aquatic Animals to Hypoxia

- Oxygen level in aquatic environment varies frequently. Aquatic animals are subjected to more frequent and rapid changes in oxygen levels.

- Many aquatic animals can withstand very long periods of hypoxia. Many of these animals utilize a variety of anaerobic metabolic pathways to survive the period of reduced oxygen availability.
- Many animals adjust the respiratory and cardiovascular systems to maintain oxygen delivery during reduced oxygen availability. For example, many fishes increase gill ventilation rate in hypoxic conditions.
- Ram ventilators (e.g. tuna) swim forward with their mouth open, increase the size of gap during hypoxia to increase water flow over the gills.

25. Panting in mammals 3 marks

Panting

- Panting a method of cooling used by many mammals, birds and some reptiles involves use of respiratory system to loss heat.
- During panting, mammals inhale through the nose and exhale through the mouth, exposing the tongue and other mouth structures to encourage further water evaporation and therefore heat loss.
- It also involves increase in breathing rate and depth (hyperventilation).
- Heat is carried away in exhalant air because the dimensions of the mouth are such that exhalant air retains the heat absorbed in the lungs.
- In contrast, nasal passages and their vascularization are effective in retaining both water and body heat.

26. Parental care 5 marks

Parental Care in Mammals

- In mammals, lactating female nourishes the young and is involved in most of the caring behaviors. Males usually play no role in raising the young.
- In some mammalian species, such as lions, males protect the females and young. In such species a male or small group of males takes care of many females at once in a harem.

27. Composition of bile 2 marks

Bile:

Vertebrate liver secretes bile. Bile doesn't contain any digestive enzymes. It contains many salts Bile salts are essential for the digestion of fats **Composition of Bile:**

Bile consists of water and a weakly basic mixture of cholesterol, lecithin, inorganic salts, organic salts, and bile pigments.

The bile organic salts are manufactured by the liver from cholesterol and amino acids complexed with sodium. The bile pigments are derived from biliverdin and bilirubin, which are products of the breakdown of hemoglobin coming from ruptured red blood cells.

28. How to nervous system maintain blood pressure 5 marks

Role of Nervous System in Maintaining Blood Pressure

Nervous control serves to maintain arterial pressure by:

- Adjusting resistance to blood flow in the peripheral circulation.

- controlling the number of capillaries open at any moment

The nervous control of capillary flow ensures that if arterial pressure falls, blood flow to the gut, liver, and muscles is reduced to maintain flow to the brain and heart.

29. Function and parts of colon 5 marks

Colon

- The colon is about 1.5 m long and consists of four parts: ascending colon, transverse colon, descending colon and sigmoid colon.
- Colon receives the undigested part of the food along with inorganic ions and excess water.
- Ions and water are absorbed by the colon.
- As water is absorbed, indigestible wastes become increasingly solid and are converted into feces as they move along the colon by peristalsis.
- It takes approximately 12-24 hours for material to travel the length of the colon and reach the rectum for elimination through the anus.

30. Vasodilation

vesodilation is the widening of blood vessels.

- o Decreased tissue oxygen levels or increased CO_2 levels.
- o Lactic acid or other acids generated by tissue cells.
- o Nitric oxide (NO) released from endothelial cells.
- o Rising concentrations of potassium ions or hydrogen ions in the interstitial fluid.
- o Elevated local temperature.

31. Which enzyme involved in pancreatic secretions

Pancreas produces enzyme-containing pancreatic juice that is delivered to the small intestine through pancreatic duct.

Pancreas produces several proteases which are secreted in the pancreatic juice.

32. What is proenzyme with example

Proenzymes

Certain digestive enzymes, in particular proteolytic enzymes, are synthesized, stored, and released in an inactive molecular form known as a proenzyme or zymogen.
Packaging of the enzyme in an inactive form prevents self-digestion of releasing tissues/cells by the enzyme. The proenzyme is activated by the removal of a portion of the molecule, either by the action of another enzyme specific for this purpose or through a rise in ambient acidity.

Examples Pepsinogen-Pepsin

Pepsin is the gastric proteolytic enzyme.

Trypsinogen-Trypsin

Trypsinogen is secreted in pancreatic juice as a proenzyme.

Chymotrypsinogen-Chymotrypsin

Chymotrypsinogen is also secreted as proenzyme in pancreatic juice.

33. Animal orientation detail 5 marks

Animal Orientation

Orientation is the ability of an animal to determine its position in space and move predictably with respect to specific stimuli. The best examples of animal orientation are observed in response to gravity. A human who has fell down on ground orients himself by standing up. Some animals orient themselves towards the incoming light. Other examples include acoustic and olfactory orientations.

Sensory Motor Basis of Orientation

Animal orientation is a complex process that requires the integration of sensory input and the coordination of motor output.

It includes receiving information through the sensory receptor neurons, processing and correlating it in the central nervous system and forming a response through the muscles that cause the body of animal to move. **Types of Orienting Movements**

1. Kinesis
2. taxis **Kinesis**

Kinesis is a change in activity or turning rate in response to a change in location.

Taxis

A taxis is an oriented movement toward (positive taxis) or away (negative taxis) from some stimulus.

34. Orthenine urea cycle 5 marks

Ornithine-Urea Cycle

- All vertebrates except most teleost fishes, synthesize urea primarily in the liver via the ornithine-urea cycle.
- In this cycle, two ammonia and one CO_2 molecule are added to ornithine to form arginine.
- Arginine is cleaved by the enzyme arginase, to remove a molecule of urea and regenerate ornithine.

35. Type of alimentary system 2

Types of Alimentary Systems:

Two basic plans of alimentary systems in animals:

A. Sac-like gastro vascular cavity
B. Tubular alimentary canal or gut

36. Baroreceptors 3 marks

Baroreceptors

The baroreceptor system is a simple and rapidly acting control mechanism that contributes to the regulation of arterial blood pressure

Baroreceptors in Vertebrates

Baroreceptors are widely distributed in the arterial system of vertebrates.

• The central cardiovascular system of amphibia, reptiles and mammals has unmyelinated baroreceptors. • These unmyelinated baroreceptors only respond to pressures above normal, initiating reflexes that reduce arterial blood pressure and thus protect the animal from damaging increases in blood pressure.

37. Three ways of survival in cellular freezing. 3 marks

Such animals employ three methods to survive freezing environments:

- Ice-nucleating Agents (INAs)
- Cryoprotectants (antifreeze substances)
- Supercooling

38. What is core temperature

The core temperatures are maintained nearly constant between 37°C and 38°C in mammals and about 40°C in birds.

39. Dose vertebrate have lungs

All vertebrate animals that live on land have lungs. When we breathe in, the muscle below the rib cage (called the diaphragm) is pulled down, and air gets sucked into the rib cage, filling the lungs. Blood cells circulating through tiny blood vessels near the lungs pick up oxygen and carry it around the body to the sites of respiration. Air is then forced out of the lungs as the diaphragm bows upwards

40. What is Rete Mirabile? Mention its function in the Tuna fish? 5 marks

Rete Mirabile

A highly evolved system of countercurrent heat exchange. Found widely in homeothermic and heterothermic vertebrates. The artery carrying warm blood toward extremity is completely encased in a circlet of veins carrying back cold blood. **Function in the Tuna fish:**

Forms an extensive heat exchange surface b/w inflowing and outflowing blood. Large number of rete mirabile in tuna (fish) are used to regulate temperature of brain, muscles, and eyes. Swimbladders of some fishes (e.g. eels) are associated with rete mirabile and functioning as CO_2 exchangers.

A countercurrent network arrangement of arterial and venous capillaries is referred to as a rete mirabile. Before entering a tissue, an artery divides into a large number of small capillaries that parallel a series of venous capillaries leaving the tissue.
The arterial capillaries are surrounded by venous capillaries.

Rete Mirabile—Functions Examples

This arrangement forms an extensive exchange surface between inflowing and outflowing blood.
Tuna (fish) have a large number of rete mirabile, which are used to regulate the temperature of the brain, muscles, and eyes.
The rete mirabile leading to the swimbladder of some fish (e.g. eels) function as a carbon dioxide countercurrent exchanger.

41. Hormones that control the pancreatic secretion

Pancreatic secretions are controlled by a variety of hormones including peptide hormones, gastrin, cholecystokinin, somatostatin and enkephalins.

The secretion of intestinal juice is regulated by several hormones, including secretin, gastric inhibitory peptide (GIP), and **gastrin**.

Gastrin secreted from the stomach lining also elicits a small flow of pancreatic juice in anticipation of the food coming in the duodenum.

42. Types of alimentary canal 2 marks

Regions of the Alimentary Canal

In general, alimentary canals can be divided on a structural and functional basis into four major divisions:

(1) **Headgut**: specialized for receiving ingested material. Includes Buccal cavity and pharynx

(2) **Foregut**: specialized for conducting, storing, and digesting ingested material. Includes esophagus and stomach. Also includes crop in some animals.

(3) **Midgut**: specialized for digesting and absorbing nutrients. Includes small intestine: duodenum, jejunum and ileum.

(4) **Hindgut**: specialized for absorbing water and defecating. Includes large intestine: colon, cecum, rectum and anus.

43. How gravity effect on arterial blood pressure 3 marks

Effect of Gravity on Arterial blood Pressure

Gravity affects the column of blood in larger arteries. So pressure in limbs and head region varies in different positions of the body.
In lying down position, the heart is at the same level as the limbs and head with respect to gravity. So, blood pressure in the arteries of head and limbs is equal.
In a standing position, the relationship between the head, heart, and limbs changes with respect to gravity. In this position heart is below the head and about a meter above the lower limbs. So, the arterial pressure in lower limbs increases while in the head region, it decreases.

Effect of Gravity on Venous Flow

Because the vascular system is elastic, an increase in absolute pressure expands blood vessels, particularly the veins which are more compliant.
Thus, pooling of blood tends to occur in veins, in different regions of the body

44. Role of nervous system in regulating blood pressure 5 marks

Role of Nervous System in Maintaining Blood Pressure

Nervous control serves to maintain arterial pressure by:

- Adjusting resistance to blood flow in the peripheral circulation.
- controlling the number of capillaries open at any moment

The nervous control of capillary flow ensures that if arterial pressure falls, blood flow to the gut, liver, and muscles is reduced to maintain flow to the brain and heart.

Nervous Innervations of Arterioles

- Most arterioles are innervated by sympathetic nerves, which release norepinephrine at their endings.
- Some arterioles are innervated by parasympathetic nerves, which release acetylcholine at their endings.

Sympathetic Nerve Stimulation

- Stimulation of sympathetic nerves causes release of norepinephrine.
- Binding of the norepinephrine to α-adrenoreceptors in the smooth muscle of arterioles causes vasoconstriction and therefore a decrease in diameter of the arterioles.

This decrease in diameter causes an increase in resistance to flow, thus reducing blood flow through that capillary bed.

45. Three local vasodilation

Vasodilation is the widening of blood vessels. It results from relaxation of smooth muscle cells within the vessel walls, in particular in the large veins, large arteries, and smaller arterioles. ...

- sildenafil (Viagra)
- tadalafil (Cialis)
- vardenafil (Levitra)

46. Chemoreceptors

chemoreceptors in our bodies, sensory cells or organs that interact with chemicals in our blood and in what we eat and smell. These chemoreceptors then convey messages about what they've found to our brains to get a response. These responses may tell us not eat something that's rotten or to flee from the smell of smoke. Our heartbeat and respiration rates are also controlled by chemoreceptors that detect carbon dioxide, oxygen, and pH levels in the blood.

Three types of chemoreceptors monitor changes in O2 and CO2 and pH levels in arterial blood:

- Peripheral Chemoreceptors
- Central Chemoreceptors
- Type-J Chemoreceptors

1. What is blubber and its role 2 marks

Blubber is a thick layer of fat, also called adipose tissue, directly under the skin of all marine mammals.

Role: As fatty tissues are little metabolically active & have little blood vascularization so it is a good insulator b/c of low thermal conductivity and prevent heat loss through body surface.

Marine mammals (seals, walruses, sirenians, cetaceans—whales) have thick layers of insulating subcutaneous fat which is known as blubber.

Blubber is a good insulator because it has a lower thermal conductivity.

Fatty tissues are metabolically very inactive and have little blood vascularization. This prevents heat loss through body surface.

2. Define Peristaltic 2 marks

Waves of contraction and relaxation in the smooth muscles present in the walls of alimentary canal. The alimentary canal starts from oral cavity and end at distal end of rectum. Esophagus conducts the swallowed bolus by peristaltic movement. The esophagus conducts the bolus by peristaltic movement.

Peristaltic movements are the waves of contraction and relaxation in the smooth muscles of alimentary canal walls.

These movements s tart from oral cavity and end at the distal end of rectum.

3. Write down the basic types of alimentary system 2.marks

Types of Alimentary Systems:

Two basic plans of alimentary systems in animals:

A. Sac-like gastro vascular cavity
B. Tubular alimentary canal or gut

4. How alveolar collapse is prevented? 2 marks

Prevention of Alveolar Collapse: Alveolar collapse normally does not occur for two reasons 1. Surrounding tissue prevents over-expansion of alveoli.

2. Presence of pulmonary surfactants.

5. Differentiate between laminar and turbulent flow? 2marks

Laminar Flow:
Flow is streamlined and continuous. The velocity of fluid is constant at every point. Flow occurs in layers at different velocities. Flow is zero at the wall and maximal at the center, A pressure difference supplies the force required to slide adjacent layers to past each other.

Turbulent Flow:
Direction of fluid movement is not aligned. It requires higher energy to move blood through Vessels. Turbulence also happens if smoothness is reduced by any obstruction e.g. Buildup of fatty deposits on arterial walls.

6. Name mechanoreceptor fiber in the arterial wall 2 marks

Atrial Receptors: Atrial walls have mechano-receptive afferent fibers including: A. Myelinated A-type afferent fibers

B. Myelinated B-type afferent fibers
C. Unmyelinated C-type afferent fibers
D. Stretch-sensitive secretory cells

7. Fahraeus- lindqvist effect 2 marks

Fahraeus-Lindqvist Effect: Phenomenon of plasma skimming & resultant increased blood flow velocity is called Fahraeus Lindqvist effect. It reduces energy required to drive blood through microcirculation

8. Types of vascular resistance 2 marks

Vascular Resistance "The resistance that must be overcome to push blood through the circulatory system and create flow"

Types

1) Systemic vascular resistance (SVR)
2) Pulmonary vascular resistance (PVR)

9. Objectives of ethology 3 marks

Ethology:
"Scientific study of animal behavior in natural environment". **Objectives of Ethology**:
Involves understanding of

1. Understanding the stimuli that elicit behavior.
2. Understanding the physiological mechanisms that mediate the response.
3. Understanding how an animal's experience during growth and development influence the response.
4. Understanding how the behavior aids in survival and reproduction.
5. Understanding the behavior's evolutionary history.

10. Four accessory glands of elementary canal? 3 marks

Accessory Glands: Secrete digestive juices through ducts.

In mammals, the alimentary canal is associated with various accessory glands that secrete digestive juices through ducts into the gut.

The accessory glands of the mammalian digestive system are:

a. Three pairs of salivary glands
b. Gastric glands
c. Pancreas
d. Liver
e. Gallbladder

11. Briefly describe ornithine urea cycle 3 marks

Ornithine-Urea Cycle

- All vertebrates except most teleost fishes, synthesize urea primarily in the liver via the ornithine-urea cycle.
- In this cycle, two ammonia and one CO_2 molecule are added to ornithine to form arginine.
- Arginine is cleaved by the enzyme arginase, to remove a molecule of urea and regenerate ornithine.

12. Factors affects systemic vascular resistance 3 marks

Factors affecting SVR

Elasticity of vessel wall

Diameter of vessel

Decreasing vessel diameter (vasoconstriction) increases SVR

Increasing vessel diameter (vasodilation) decreases SVR

Vascular resisance is determined mainly by the muscle tone in the smooth muscle tissue of the tunica media and the elasticity of the elastic fibers there.

Vasoconstriction (i.e., decrease in blood vessel diameter) increases SVR, whereas vasodilation (increase in diameter) decreases SVR.

13. 3 uses of crop in various animals. (3) marks

Crop

- A sac-like expanded section in esophagus of some animals. Used to store food without digesting. Crop plays various roles in different animals. For example
- Food storage without digestion
- Fermenting the food
- Prepare food for nestlings (pigeon's milk)

In some animals crops are also used to ferment or digest foods for purposes other than normal digestive process. For example some parent birds prepare food in this way to be regurgitated for their nestlings e.g. pigeon's milk.

14. Name 3 vasodilators in the body? 3 marks

1. Histamine released from connective tissue and white blood cells in injured tissues causes vasodilation
2. Plasma kinins (e.g. bradykinin) are potent vasodilators in damaged tissues
3. Nitric oxide (NO) serves as a major inducer of vasodilation in cardiovascular system also causes relaxation of vascular smooth muscle and acts as vasodilator
4. Lactic acid generated by tissue cells
5. Rising conc. of K+ or H+ ions in interstitial fluid
6. Prostacyclin causes vasodilation and acts as an anticoagulant.

15. Write the name of local vasoconstrictor? 3 marks

1. Prosta gland and thromboxanes released by activated platelets and white blood cells.
2. Endotheline released by damaged endothelial cells.

If autoregulation fails to normalize conditions at the tissue level, neural mechanisms and endocrine factors are activated.

16. Write down the composition of the Bile? 3 marks

Bile: Vertebrate liver secretes bile. Bile doesn't contain any digestive enzymes It contains many salts Bile salts are essential for the digestion of fats **Composition of Bile:**

Bile consists of water and a weakly basic mixture of cholesterol, lecithin, inorganic salts, organic salts, and bile pigments.

The bile organic salts are manufactured by the liver from cholesterol and amino acids complexed with sodium.

The bile pigments are derived from biliverdin and bilirubin, which are products of the breakdown of hemoglobin coming from ruptured red blood cells.

17. What is Rete Mirabile? Mention its function in the Tuna fish? 5 marks

Rete Mirabile A highly evolved system of countercurrent heat exchange. Found widely in homeothermic and heterothermic vertebrates. The artery carrying warm blood toward extremity is completely encased in a circlet of veins carrying back cold blood.

Function in the Tuna fish:

Forms an extensive heat exchange surface b/w inflowing and outflowing blood. Large number of

rete mirabile in tuna (fish) are used to regulate temperature of brain, muscles, and eyes. Swimbladders of some fishes (e.g. eels) are associated with rete mirabile and functioning as CO_2 exchangers.

A countercurrent network arrangement of arterial and venous capillaries is referred to as a rete mirabile. Before entering a tissue, an artery divides into a large number of small capillaries that parallel a series of venous capillaries leaving the tissue.

The arterial capillaries are surrounded by venous capillaries.

Rete Mirabile—Functions

This arrangement forms an extensive exchange surface between inflowing and outflowing blood.

Examples

Tuna (fish) have a large number of rete mirabile, which are used to regulate the temperature of the brain, muscles, and eyes.

The rete mirabile leading to the swimbladder of some fish (e.g. eels) function as a carbon dioxide countercurrent exchanger.

18. Note on physiological adaptation of ectothermic in hot environment 5 marks

Physiological Adaptations

- Ectotherms have high heat conductance
- Radiate heat rapidly—by moving to less warmer places
- Panting: Above 40oC many reptiles start panting—like birds and mammals
- Increases heat loss through respiratory evaporation, causing cooling.
- Regulating heart rate and flow of blood to surface tissues—to eliminate heat (e.g. marine iguanas)
- Gila monster (Heloderma suspectum) evaporates water from its cloaca to cool—a process similar to sweating in mammals

Most ectotherms have high heat conductance that allows them to radiate heat rapidly on moving to less warm places.

Above 40oC many reptiles start panting, just like birds and mammals to increase heat loss through respiratory evaporation that causes cooling.

The marine iguanas can regulate heart rate and flow of blood to the surface tissues to either absorb or eliminate heat.

The Gila monster (Heloderma suspectum) evaporates water from its cloaca to cool its internal body temperature—a process similar to sweating in mammals.

19. Write a note on types of stomach in vertebrates 5 marks

Types of Stomach in Vertebrates: Based on number of chambers, stomachs are classified as: a. Monogastric stomachs

b. Digastric stomachs **Monogastric Stomachs**

A monogastric stomach consists of a single strong muscular tube or sac.

Carnivorous and omnivorous vertebrates have a monogastric stomach.

Digastric Stomachs

Digastric stomachs are multichambered and are found in ruminant mammals of order Artiodactyla e.g. deer, giraffe, sheep, cattle, camel, llama etc.

These stomachs have four chambers which are separated into two divisions.

- The first division consists of fermentation chambers called rumen and reticulum.
- The second division comprises the omasum and abomasum (true stomach).

20. What is hypoxia? Responses of aquatic animals in it 5 marks

Hypoxia • "A condition in which the body or a region of the body is deprived of adequate oxygen supply" **Hypoxia in Aquatic Environment**

- O2 levels in aquatic environment vary more frequently
- Aquatic animals are faced with frequent and rapid changes in oxygen levels. Many aquatic animals can withstand very long periods of hypoxia **Respiratory Responses of Aquatic Animals to Hypoxia**

Oxygen level in aquatic environment varies frequently. Aquatic animals are subjected to more frequent and rapid changes in oxygen levels.

Many aquatic animals can withstand very long periods of hypoxia. Many of these animals utilize a variety of anaerobic metabolic pathways to survive the period of reduced oxygen availability.

Many animals adjust the respiratory and cardiovascular systems to maintain oxygen delivery during reduced oxygen availability. For example, many fishes increase gill ventilation rate in hypoxic conditions. Ram ventilators (e.g. tuna) swim forward with their mouth open, increase the size of gap during hypoxia to increase water flow over the gills.

21. What are pheromones? Also describe their Role? 10 marks

Pheromones:

- Chemical messengers released by many animals in environment to communicate. Spread through air.

Characteristics:

a. Affect the behavior of individuals of a species
b. Effective at remarkably low concentrations
c. Have particular odors
d. Detected by olfactory receptors of receiving animal **Roles of Pheromones:**

1. Recognize Members of Species:
Unicellular organisms recognize members of their own species. Many insects identify members of their colony.

2. Reproductive Roles:
Pheromones attract members of opposite sex for breeding.

Induce courtship behaviors in members of a species.

Induce spawning behavior in aquatic animals with external fertilization.

3. Territorial Behaviors:
Many male mammals mark their territories with pheromones. It warns other males of their occupied area. **4. Insect's Social Order:**

In a honeybee colony, pheromones produced by queen maintain hive's complex social order.

5. Alarm Calls:
Members of some species produce pheromones on getting injured.

It warns other members by the presence of danger in area.

6. Repel Predators:
Pheromones produce by the pray may repel predators.

Example: The skunks produce foul-smelling musk that makes skunks unpalatable or distasteful to their enemies.

22.Define vestibular apparatus ? 2marks

Vertebrate Organ of Equilibrium

- Called as vestibular apparatus, located in the inner ear. It has:

1. • Saccule
2. • Utricle
3. • Semicircular canals

23.Abnormalities due to GH? 5 marks

Disturbances in the secretion of GH lead to abnormal growth and development in humans:

Gigantism

- Hypersecretion of GH before puberty causes excessive size and stature.

Developmental Abnormalities Due to GH

Acromegaly

- Hyper secretion of GH after maturity causes enlargement of bones of the head and limbs.

Dwarfism

- Insufficient secretion of GH during childhood and adolescence causes underdevelopment of The body (short stature).

24.Oxytocin? 3 marks

Is a type of harmone . oxytocin are released by axon terminals in the posterior pituitary. Stimulates milk ejection, uterine contractions during child birth.

Neurohypophyseal hormones are synthesized and packaged in the cell bodies of two groups of neurosecretory cells in the anterior portion of the hypothalamus:

1. Supraoptic nuclei
2. Paraventricular nuclei

25. Five types of hormone ?

Based on their structure and pathway for synthesis, hormones are divided into four groups:

- Peptide and protein hormones
- Amine hormones
- Steroid hormones
- Prostaglandins

26.Gastrointestinal track? 5

Gastrointestinal tract

Cholecystokinin Stimulates secretion of enzymes by pancreatic acinar cells

Stimulates gall bladder contraction

Chymodenin Stimulates secretion of chymotrypsinogen from pancreas

Gastrin Stimulates HCl secretion in stomach

Secretin Stimulates secretion of bicarbonate

Substance P Acts as enteric neurotransmitter for feeling of pain **Motilin** Increases motility of intestinal villi

27. Prostaglandins.10 marks

Prostaglandins are about 16 hormone-like local regulators substances that constitute a family of cyclic, long-chain, unsaturated, hydroxy fatty acids.

Synthesis

Prostaglandins are produced by all or nearly all tissues. They are synthesized in membranes from arachidonic acid, which is produced by cleavage of membrane phospholipids by phospholipases.

Mode of Action

In some cases, they act locally as paracrine agents and in other cases they act on distant target tissues in endocrine fashion. They have a rapid, short-lasting effect, similar to that of lipid-insoluble hormones.

They bind to cell-surface receptors linked to the cAMP pathway.

Functions

Prostaglandins have diverse actions on a variety of tissues, particularly involving smooth muscles.

1. Role in Fertilization

Prostaglandins present in semen aid in fertilization by reacting with the female cervical mucus to make it more receptive to sperm movement. They also stimulate the smooth muscles of the female's uterine wall to contract, helping sperm reach an egg.

2. Aid During Labor

Certain prostaglandins are secreted by the placenta at the onset of childbirth. They make the muscles of uterus more excitable, enhancing uterine contractions during labor.

3. Role in Immune Response

The damaged tissues, that induce immune response, produce prostaglandins. Some prostaglandins act as local regulators of inflammation, promoting fever and inflammation and also intensify the sensation of pain. They worsen pain by increasing nociceptor sensitivity to noxious stimuli. The interleukin-1, produced in immune response, causes fever by inducing the formation of a prostaglandin E2 which acts in the hypothalamus to

elicit the fever reaction. The anti-pyretic, anti-inflammatory and pain-relieving effects of aspirin and ibuprofen are actually due to the inhibition of prostaglandin synthesis by these drugs.

4. Blood Clot Formation

Prostaglandins also help the platelets to aggregate and form blood clots. This is the basis of use of aspirin by patients at risk for a heart attack due to formation of clots.

5. Stomach Lining

Prostaglandins also help maintain a protective lining in the stomach.As aspirin interferes with prostaglandin synthesis, long-term aspirin therapy can result in damage to this protective lining, causing debilitating stomach irritation.

6. Production of Erythropoietin

Several prostaglandins stimulate the production of erythropoietin by kidneys. Erythropoietin is a hormone that stimulates the production of erythrocytes.

28.neuromodulation?

A physiological process by which neurotransmitters released by a small group of neurons diffuse through large areas of the CNS and interact and modulate the effect of many neurons simultaneously.

- This is in contrast to the normal synaptic transmission. Neurotransmitters involved in altering the cellular properties and efficacy of synaptic transmission of multiple postsynaptic neurons.
- They act through Gprotein linked receptors that use voltage-gated ion channels for ion movements.
- Generation of response is slow, but lasts longer.

29.role of corpus leutum in pregnancy

If the egg is fertilized and implants, active corpus luteum is maintained.

- So progesterone and estrogen secretion continues until placenta fully develops to take over the production of these hormones.
- At this stage, corpus luteum degenerates. In many mammals, (e.g. rat) corpus luteum is stimulated by prolactin and continues to grow and secrete estrogen and Progesterone throughout the gestation period.

30.Types of stimuli

Two types:

- External stimuli Stimuli of odor, touch, light, sound and gravitation
- Internal stimuli Stimuli of pain, homeostatic imbalances and blood pressure

31.taste buds and taste receptors

The gustatory organs ofvertebrates are called taste buds.

A taste bud is composed of about 50 types of modified epithelial cells, including:

- Supporting cells (sustentacular cells)

- Basal cells
- Taste receptor cells
- basal cells are progenitor cells that give rise to new
- They regularly generate new sensory taste receptor cells which have an active life of only 10 days.

32.Feedback Mechanism (2)

Mechanism of self-regulation of Biological processes. Output or product itself regulates the process Maintenance of homeostasis

33.Acclimatization with example (2)

Acclimatization is a physiological, biochemical, or anatomic change within an individual animal that results from the animal's chronic exposure to new, naturally occurring environmental conditions. Example.

Let's consider an animal that voluntarily migrates from a valley to a high mountain i.e. a voluntary change happens in its natural environment in which oxygen partial pressure is low.

34.Adrenal cortex hormone (3)

Adrenal (Cortex)

Aldosterone Increases blood Na+levels; increase K+secretion

Cortisol, corticosterone, cortisone

Role in carbohydrate metabolism increase blood glucose levels anti-inflammatory effects

35.Parts of eye take part in focusing (3)

The parts of the eye involved in focusing and image formation are:

- Cornea, a biconvex lens, pupil and retina.

36.Role of T3 and T4.10 marks

Metabolic Roles of T3 & T4

- They regulate bioenergetics by stimulating cellular respiration, oxygen consumption and metabolic rate.
- The acceleration of metabolism leads to a rise in heat production. This is of major importance in the thermoregulation of many vertebrates.
- They also sensitize some tissues to epinephrine that helps to maintain normal blood pressure, heart rate and muscle tone.
- They also promote normal motility of gastrointestinal tract and regulate reproductive functions.

37.Acclimatization2 marks

An adaptive change within an individual animal.

- Happens due to chronic exposure to new, naturally occurring environmental conditions.

38.Rhodopsin 2 marks

The visual pigment in the rods is rhodopsin. Rhodopsins absorb photons of light energy and produce a generator potential.

Opsins are coupled to light-absorbing photopigment Retinal.

- This form a functional pigment molecule rhodopsin.

39.define line coding 2 marks

Each receptor subtype for taste sensations is connected to a particular set of axons in the nerve.

- In such an arrangement, information about one taste e.g. "sweetness" would be carried by some specific subset of axons.
- Such a pattern is called labeled line coding.

40. Explain the types of hair cells

Hair cells are ciliary cells.

- Found in several sensory organs of vertebrates.
- They are highly sensitive mechanoreceptors.
- Responsible for transducing mechanical stimuli into electrical signals.

Systems Based on Hair Cells: Examples

- Lateral-line system of fishes and amphibians, involved in detection of motion in the surrounding water.
- This system is based on hair cells.

Hair Cell Cilia

- Many cilia project from the apical end of each cell (reason of naming).
- Cilia are of two types:
- Kinocilium • Stereocilia
- A hair cell has a single kinocilium.
- Kinocilium has a "9 + 2" arrangement of internal microtubules (similar to other motile cilia).
- Each hair cell has 20-300nonmotile stereocilia.
- Stereocilia are structurally and developmentally distinct from the kinocilium.
- They are not formed of microtubules.
- They are formed of actin filaments.
- The stereocilia are arranged in order of increasing length from one side of the cell to the other.

41.diseases by hypothyroidism

Characterized by low production of throxine.

- It results from the lack of dietary iodine.
- It is represented by two types of diseases:
- Cretinism
- Goiter
- Cretinism results from iodine deficiency during early stages of development. In cretinism somatic, neural and sexual development is severely retarded, metabolic rate and resistance to infection is also reduced.
- Inadequate production of thyroxine in adults leads to goiter. In goiter, TSH is produced excessively that causes overstimulation of thyroid gland resulting in its enlargement (hypertrophy).

42.Name glandular cells of thyriod and name their hormones too. (3marks)

Glandular Cells and Hormones of Adenohypophysis

1. Somatotropes—produce growth hormone (GH)
2. Corticotropes—produce adrenocorticotropic hormone (ACTH)
3. Thyrotropes—produce thyroid-stimulating hormone (TSH)
4. Gonadotropes—produce gonadotropic hormones (luteinizing hormone, LH and follicle stimulating hormone, FSH)
5. Lactotropes—produce prolactin (PRL)

43.Range fractionation

In a sense organ, sensory receptors are arranged in an order of increasing sensitivity to different range of intensities of the stimulus.

- This hierarchical arrangement of receptors is known as range fractionation. Each individual receptor, in a sense organ, covers only a fraction of the total dynamic range of the stimulus intensities.
- Receptors work together in a hierarchical way to provide discrimination of stimulus intensities.

44.Placental hormone

In pregnancy, placenta secretes large quantities of hormones, essential for maintenance of pregnancy.

- These hormones include:
- Human chorionic gonadotropin
- Estrogens
- Progesterone
- Human chorionic somatomammotropin

45.Progesteron

Progesterone is secreted in tremendous quantities by placenta.

- This secretion is about 10-fold more than normal levels. Causes development of decidual cells in endometrium. These cells play an important role in the nutrition of the early embryo.
- It decreases contractility of the uterus to prevent spontaneous abortion.

46.Color blindness 3

Color blindness is caused due to a mutation in one of the cone opsin genes that results in the absence of one type of pigmented cones.

- A person missing a single type of color receptive cones is unable to distinguish some colors.

47.Three use of crop?

A sac-like expanded section in esophagus of some animals. Used to store food without digesting. • Generally found in infrequent or quick feeders • Allows quantities of food to be stored for digestion at a later time

48.Define peristaltic?

The wavelike muscular contractions in tubular structures, especially organs of the digestive system such as the esophagus and the intestines. **Peristalsis** is characterized by alternate contraction and relaxation, which pushes ingested food through the digestive tract towards its release at the anus.

49.Objective of ethology?

Ethology is the scientific and **objective** study of animal behaviour, usually with a focus on behaviour under natural conditions, and viewing behaviour as an evolutionarily adaptive trait.

50.Three local vasodilator?

Vasodilation is the widening of blood vessels. It results from relaxation of smooth muscle cells within the vessel walls, in particular in the large veins, large arteries, and smaller arterioles. ...

- sildenafil (Viagra) tadalafil
- (Cialis) vardenafil (Levitra)

51.Three respiratory pigments?

A **respiratory pigment** is a molecule, such as hemoglobin in humans and other vertebrates, that increases the oxygen-carrying capacity of the blood. The four most common invertebrate **respiratory pigments** are hemoglobin, hemocyanin, hemerythrin and chlorocruorin.

52.Differentiate between Norepinephrine and Epinephrine?2

Norepinephrine is the primary excitatory transmitter in postganglionic cells of sympathetic system. Epinephrine is excitatory at some synapses and inhibitory at others depending on postsynaptic membrane

53.Define vestibular apparatus?

The vertebrate organ of equilibrium is the vestibular apparatus. Vestibular apparatus, located in the inner ear. It has:

..The saccule

- The utricle
- Three semicircular canals lying orthogonally in three mutually perpendicular planes.

54.Name two components of rhodopsin ? 2mks

A rhodopsin molecule consists of two major components:

Opsin protein

A light-absorbing molecule (retinal or 3dehydroretinal) Function of LH in male and female.

In Females LH stimulates production of estrogen and progesterone by ovary.

It causes ovulation and formation of corpus luteum.

In males, LH is also called as ICSH (interstitial cell stimulating hormone) It stimulates testosterone production by the Leydig cells of testes.

55.What are gonadotropin

Gonadotropins: Luteinizing and Follicle Stimulating Hormones. Luteinizinghormone (LH) and folliclestimulating hormone (FSH) are called gonadotropinsbecause stimulate the gonads - in males, the testes, and in females, the ovaries.

56.Define the terms gland and secretion? 2mks

A gland is a cell or group of cells that secretes a particular chemical substance for use in the body or for discharge into the surroundings.

Secretions are chemical substances, synthesized by glandular cells. They are released from the gland in response to an appropriate stimulus.

57.Write the role of Fibers of Zonula and ciliary muscles in eye? 3

Fibers of zonula held the lens in place. They change the shape of lens by exerting an outward tension on perimeter. Attached with fibers of zonula are the ciliary muscles
They contract to pull fibers of zonula, which flatten the lens. This focuses distant objects on retina. When they relax, lens becomes rounded and near object is focused.

58.Describe Hodgkin cycle in detail? 3

The number of Na+ channels open at any instant depends on Vm as well as on time i.e. the phase

of action potential. Thus changes in the conductance of sodium ions (gNa) occur as a function of Vm and time and reflect the behavior of thousands of Na+ channels, each one opening and closing during depolarization in accord with certain principles. During an AP, Na+ channels respond to an initial depolarization by opening, allowing Na+to enter the cell, which further depolarize the membrane. This depolarization causes more channels to open, allowing still more Na+to enter the cell and triggering an explosive, regenerative event. This relationship between membrane potential and sodium conductance is termed as the Hodgkin cycle and represents a type of positive-feedback system.

A stimulus causes local depolarization of the membrane. Na+ channels open in response to initial depolarization. This increases Na+ conductance that allows Na+ to enter the cell, which cause further depolarization of membrane and decrease in Vm. This relationship between Na+ conductance (gNa) and Vm represents a type of positive feedback system and is termed as the Hodgkin cycle.

It results in triggering of an action potential.

59.Which hormones are involved in the regulation of water and ions? 3

These hormones include:

- Antidiuretic hormone (ADH)
- Aldosterone
- Atrial natriuretic Hormone
- Calcitonin
- Parathormone

60.Messiner corpuscales and paccin corpuscales difference?3

Touch receptors in ridges of fingertips are Meissner's Corpuscles. They have encapsulated nerve endings. To receive pressure stimulus, Pacinian corpuscles are situated deep in skin, in the limbs they receive vibrations.

61.Basic components of homeostasis and example of homeostasis, Its mechanism (5max) Examples of Homeostasis:
Homeostatic regulation of water • Homeostatic regulation of temperature • Regulation of pH • Regulation of glucose concentration • Regulation of osmotic pressure • Regulation of oxygen level • Regulation of ion conc.

Mechanism

Variables

Set point

Living control systems VS physical control systems

Three components: Receptors, Control centre, effectors

Example

• Temperature control system in air conditioners and water heating geysers VS Endothermic animals

62.Name glandular cells of thyriod and name their hormones too. (3mqx)

It contains five types of glandular cells that synthesize and secrete six hormones

1-Somatotropes produce growth hormone (GH) **2-Corticotropes produce adrenocorticotropic hormone** (ACTH) **3-Thyrotropes**
Produce thyroid-stimulating hormone (TSH)

4-Gonadotropes
Produce gonadotropic hormones (luteinizing hormone, LH and follicle stimulating hormone, FSH)

5-Lactotropes
Produce prolactin (PRL)

63.Exocirne gland with 2 examples atleast (5max)

Exocrine glands produce fluid secretions that are delivered through ducts onto the epithelial surfaces of the body. The fluid secretions may be proteins (enzymes) or mucous or both.

Examples:

Salivary glands produce saliva that is delivered to the oral cavity through submandibular and parotid ducts.

Pancreas produces enzyme-containing pancreatic juice that is delivered to the small intestine through pancreatic duct.

Lacrimal glands produce tears that is delivered through lacrimal duct on the surface of eye.

Mammary glands produce milk that is delivered through lactiferous ducts to the nipples

Androgens secreted by adrenal cortex 2 marks Androgens secreted by adrenal cortex include:

1. Testosterone
2. Dihydrotestosterone
3. Androstenedione
4. Dehydroepiandrosterone (DHEA).

64.Effects of neuro-modulation 3 marks

- Development of complex behavioral patterns e.g. happiness, exploration, revenge, reward, greed.
- Processes of thinking, cognition, planning, learning and memory • Behavioral problems: mood swings, sleep disturbances, feelings of stress, anxiety, anger and depression.

65. 2 examples of interoceptive receptors 3 marks

These include:

Receptors of vestibular system that monitor orientation of the body

Thermo-receptors that keep track of thermal state of the body

Chemo-receptors that keep track of chemical state of the body Proprio-ceptors that monitor position of the muscles and joints Nociceptors perceive the sensation of pain. Pacinian corpuscles receive pressure stimulus.

66.Name and explain hair cell types 5 marks

Inner hair cells (IHC)

There are about 3,500 IHCs arranged in a straight line or wide U. They are true sensory cells which send impulses via auditory nerve.

Outer hair cells (OHC)

About 12,000-25,000 OHCs in mammals, they are arranged in three or four rows forming a characteristic W shape.

Outer hair cells have both sensory and motor elements. They contribute to hearing sensitivity and frequency selectivity

67.Hormones of ovary

LH stimulates production of estrogen and progesterone by ovary.

68.Explain photoreceptor

Photoreceptors can detect and generate response to a single photon of light. Photoreceptors possess lightsensitive carotenoid pigments retinal and 3dehydroretinal. Carotenoids are associated with opsin proteins to form rhodopsins. Rhodopsins absorb photons of light energy and produce a generator potential

69.Hair cilia types

Many cilia project from the apical end of each cell (reason of naming). • Cilia are of two types:

• Kinocilium • Stereocilia

A hair cell has a single kinocilium. • Kinocilium has a "9 + 2" arrangement of internal microtubules (similar to other motile cilia). • Each hair cell has 20-300nonmotile stereocilia. • Stereocilia are structurally and developmentally distinct from the kinocilium. • They are not formed of microtubules. • They are formed of actin filaments. • The stereocilia are arranged in order of increasing length from one side of the cell to the other.

70.Diseases by hypothyroidism

It is represented by two types of diseases: • Cretinism • Goiter

Cretinism results from iodine deficiency during early stages of development Inadequate production of thyroxine in adults leads to goiter.

71.active potential phases Stigma and ocelli 3

Eyespot or Stigma

Simplest photoreceptive structure. Found in some protozoa e.g. euglena. It is a bright red colored organelle and has carotenoid pigments. It gives a sense of light and dark. Helps in phototaxis.

Eyecups or Ocelli

Multi-cellular photoreceptive structure that consists of a cuplike depression containing photoreceptor cells; Found in cnidarians and flatworms e.g. Planaria.

Cannot form image, provides the animal a sense of direction only

72Functions of T3 and T4 and Hyperthyroidism 10. marks

These hormones have diverse effects on the physiology of virtually all tissues of the body. Their roles can be grouped into two categories:

1. Metabolic roles
2. Developmental roles

Metabolic Roles of T3 & T4
They stimulate cellular respiration, oxygen consumption and metabolic rate and, in turn, heat production. (important in thermoregulation).

They sensitize some tissues to epinephrine that helps to maintain normal blood pressure, heart rate and muscle tone.

They also promote normal motility of gastrointestinal tract.

They also regulate reproductive functions.

Developmental Roles of T3 & T4
They significantly affect the development and maturation of vertebrate animals.

They are involved in the normal functioning of bone-forming cells and the branching of nerve cells during embryonic development of the brain.

They control metamorphosis of a tadpole larva into adult frog.

Developmental effects of growth hormone occur only in the presence of thyroid hormones.

Hyperthyroidism
Excessive secretion of thyroid hormone is known as hyperthyroidism.

The most common form of hyperthyroidism is Graves' disease. It is an autoimmune disorder in which antibodies that

mimic TSH bind to the receptor for TSH and cause sustained thyroxine production.

It leads to high body temperature, profuse sweating, weight loss, irritability, high blood pressure and protruding eyes

(exophthalmia).

BEST OF LUCK

BEST OF LUCK

ZOO502-Animal-Physiology-and-Behavior MCQs FINAL TERM (PAST PAPERS)

1. Non diffusible ___produce Donnan effect _____**SOLUTE**
2. ____ channels are involved in producing action potential _____**Na+ and K+**
3. AChE abundantly present in _____**SNYPTIC CLEFT**
4. Which ___monitors the position of muscles and joints _____**PROPRIOCEPTORS**
5. Sensory sensilla are organ of ___in insects _____**TASTE**
6. Merkel Disks are associated with the reception of _____**VIBRATION**
7. Human ear can detect sound _____**20 TO 20000 HRTZ**
8. interstial tissues constitute___ of adult testes mass_____**20%** 9. Adenohypophysis is the site for production of _____ options?
10. Velocity of impulse conducting by arterial fiber is _____**0.3 m/s**
11. Equation indicate blood flow _____**V = Q/A**
12. Relative viscosity of plasma _____**1.8**
14. Fenestrated capillaries are found in _____ **endocrine glands**
15. Which is local vasoconstrictor _____**Angiotensin II**
16. ___ is formed in lungs, acts as local vasoconstrictor _____**Angiotensin II**
17. Amount of air M/V that still remains in lungs when they fully collapse _____**30 to 120mL**
18. Respiratory system of ___ is composed of 7 or 9 air sacs_____ **Birds____**
19. Central chemoreceptors are located in _____**Medulla of mammals and Air breathing vertebrates**
20. which are stenohaline animals_____ **Marine spider, crab**
21. A camel can lose ___ of water and still survive _____**40%**
22. ___ is involve in maintaining pH of urine_____ **TUBULAR SECRATION**
23. GFR averages an astounding ___ per minute_____**125 mL**
24. It requires only __ of water to excrete 1g of urea._____ **0.05 L**
26. Esophagus is included in ___ division of mammalian digestive tract_____**Four (4)**
27. Human GIT secrets__ exocrine fluid daily _____**Bile and Bile salts**

28. Goblet cells synthesis and secretes_____ **Mucus**
29. ___ is largest water soluble essential nutrient _____**Vitamin B12**
30. Brown fat contains large amount of _____**Mitochondria**
31. Which is Neuroethological behavior _____**Study behavior in laboratory**
33. Agonistic behavior involve___b/w animals_____ **Competition**
34. Social behaviors are based on _____ **Competition**
35. Respiratory epithelium has only a thickness of _____**0.5 to 15 um**
36. High pressure is _____ **Systolic BP**
37. Atrial natriuretic peptide (ANP) hormone secreted by _____**Stretch-Sensitive Secretory Cell**
38. Blood with hemoglobin carry ___ Oxygen per 100ml._____**20 ml**
39. Inspiratory volume is _____**1900 ml**
40. DCT secrete _____ **NH4+ ions, K+ ions, & H+ ions.**
41. Teloest fish is _____**Ammonotelic Animals**
42. Esophagus is part of _____**Foregut**
43. Deudenum is _____cm long _____**25cm**
44. Troponin is sensitive to _____ **Ca+**
45. Electric impulse is observed in _____**Cardiac muscle**
46. Proteases are _____(**trypsin , chymotrpsin , pepsin**)
47. Facilitated diffusion occurs with _____**channel proteins** 48. Sweating occur in _____**UCT (Upper Critical Temperature**)
49. Thickness of muscle fibers is ____?
50. 1 joule of energy utilized in work, _____energy is degraded_____**3Joule**
51. Study behavior in laboratory _____**Neuroethological**
52. A honey bee finds a food source close to the _____**50meter**
53. One female has an exclusive relationship with two or more males' _____ **poly androus**
54. Social behavior _____(**Cooperation , competition, conflict**) 55. Agonistic behavior_____ (**threat display, attach and fight**)
56. which is not related to physiology._____**None**.

57. pineal glands are found in dorsal surface of ..... in vertbrates_____ **forebrain**.

58. insp3 releases the ca2+ ions from intracellulr calcium stores like...

A.. mitochondria and endo plasmic reticulum

B... golgi appratus and ER

C.... ER and ribosomes

D.. mitochondria and golgi appratus

59. ca2+ flow out of lumen of SR increases free ca2+ Conc. In myoplasm to an active level of about..___**10^{-7}M**

5. human S-A node is ..... thick_____**1mm**

6. Rhythmical discharge of sinstric node is..... per minute._____**70 to 80**

7. females IRV is ....._____**1900 ml**

8. related to emesis_____**reverse peristalsis**

9. Velocity of impulse conducting by arterial fiber is_____ **0.3%**

10. The relative viscosity of the plasma _____ **1.8**

11. Fenestrated capillaries are found in _____ **Endocrine gland**

12. . _____ is the local vasoconstrictor._____**Angiotensin II**

13. Respiratory system of _____is composed of 7 or 9 air sacs. _____**Birds**

14. . A camel can lose _____ of water and still survive. _____**40%**

15. GFR averages an astounding _____ per minute._____ **125ml**

16. Esophagus is included in _____ division of mammalian digestive tract. _____**Four**

17. Brown fat contains large amount of _____ **Mitochondria**

18. It requires only ____ of water to excrete 1g of urea._____ **0.05 L**

19. Which is Neuroethological Study behavior in laboratory_____ **There are no predators**

20. Esophagus is the part of _____**Foregut**

21. Troponin is sensitive to _____**Ca+**

22. 1 joule of energy utilized in work, _____ energy is degraded. _____**3 J**

23. A honey bee finds a food source close to the _____**50m**

24. One female has an exclusive relationship with two or more males, _____**Polyandrous**

25. Social behavior _____. (cooperation, conflict, competition)_____**All**

26. Agonistic behavior _____. (Threat display, attach, fight)_____ **All**
27. The conc. of K-ion is maintained _____ times more in the cytosol than extracellular fluid._____ **10-30**
28. A depolarization in the range of _____ is the threshold that triggers an AP. _____**-55 to -30**
29. Meissner's corpuscles receptors are for _____ **Touch**
30. Which one of the following is not local vasoconstrictor_____**Endorphins**
31. Example of peptide hormone_____ **Oxytocin**
32. Cytosolic conc. of calcium ions remains _____ **Below 10-6**
33. _____ that induce hallucinations._____ **Mascalline**
34. Example of steroid hormone is _____ **Aldosterone**
35. Adenohypophysis contains_____ types of glandular cells that synthesized and secrete hormones._____ **5**
36. Melatonin belongs to _____ **Pineal gland**
37. _____ releases Ca-ions from bones to blood. _____**Parathormone**
38. Sweating occurs by _____ **UCT upper critical temperature**
39. The study of physics law in physiology of _____ is called as hemodynamics._____ **Circulatory**
40. _____ involves maintaining the pH of urine._____ **Tubular reabsorption**
41. . Epithelium layer of the small intestine produce the enzyme. _____**Pepsin**
42. Example of temporal heterotherm _____ **Python**
43. which of these is temporal heterotherms _____**humming bird**
44. luminal membrane of intestinal epithelial cells have …….. types of transport protein_____**Five**
45. human GIT secrete …….. litre of exocrine fluid per day_____**3**
46. ……. recieves bile salt from gall bladder._____ **Duodenum**
47. esophagus is icluded in _____**foregut**
48. length of a muscular twich is _____**10msec**
49. equation for blood flow is_____**V=Q/A**
50. warmth receptors has conduction velocity _____**19m/s**
51. which of the following in related to ATPase_____ **Association with Na+/K+ pumps**
52. Which of the following in not related to voltage gated sodium channels _____**Slightly smaller than the potassium channels**

53. AChE is abundantly present in_____ **Synaptic cleft**

54. Human have more than----------different types of recepters for bitter taste_____**30**

55. Markels disks are associated with reception of _____ **vibration**

56. Cold recepters are------ more common in skin than heat receptors._____**3.5 times**

57. Melatonin belongs to_____ **Pineal gland**

58. In ---- protectin regulates fat metabolism and reproduction_____**Birds**

59. In the synthesis of catecholamines conversion of tyrosine to dopa and dopamine occure in the_____ **Cytosol**

60. Thyroglobuline is stored within the thyroid gland in_____**Follicles**

61. Hypothyrodism leads to ---_____**- a) High body temp. b) Cretinism c) Weight loss d) Profuse sweating**
?

62. When the force exerted by the muscles contraction is equal to the opposing external force then ----- contractions occur_____ _____**Isometric**

63. In equation R= , stands for _____**Resistance to blood flow**

64. Atria blood pressure is highest during_____ **Ventricular systole**

65. Blood with hemoglobin can carry ----O2 per_____**20 ml**

66. Central chemoreceptor are located in_____ **Medulla of mammals**

67. which of the following produces atrial natriuretic peptide (ANP) hormone._____**strech sensitive secretor cells**.

68. which of the following is an example of negative feedback mechanism _____**control of blood sugar by insulin**

69.Which of the following is related to stenohaline____ **they can carry osmotic regulation in face of changing salinity**

70. ----is involved in maintaing ph of urine_____**tubular secreation.**

71. Structure and function level exists at, (system-muscle, cell-muscle, macromoleculesarcomere)_____**All**

72. Electric transmission in cells has observed in _____--_**smooth muscles fibers, cadic muscle fiber**.

73. Synaptic cleft size is (1mm, 2mm, 3mm, 4mm)? _____ **20nm wide**

74. Acromatic vision is due to_____**rods**.

75. Secretions that affect neighboring cells are_____**Paracrine**.

76. Incorrect statement about receptors of taste_____(**salty taste is associated with ions**)

77. Hypothalamic releasing hormones except_____**thyrotropin**

78. . Myofibril diameter is_____**1-2um.**

79. Produces ANP hormone_____ **striated sensitive fibers**

80. Resting tidal volume_____ **Averages about 500 mL in human male and female both**

81. Statement relevant to stenohaline_____**animals that are found in estuary and river-sea water coonfluence**

82. GFR average is_____ **125ml**

83. Length of jejunum is _____**2.5m**

84. Salivary secretions have_____**HCO3**

85. Epithelium of small intestine produces enzyme_____ **proteolytic enzyme**

86. Laminal membrane of small intestine-----types of transport protein_____ (**4,5,6,7**)

87. Relevant instinctive behavior development after exposure to environment _____ (**agonistic, social, territorial adaptive**) ?

88. Neurohypophysis is .....lobe of pituitary gland_____**posterior**.

89. Receptor of thyroid gland are found in .....

90. Brow fate contain....as appear brown_____**large number of mitochondria.**

91. Blood with hemoglobin can caery.....oxygen is_____**20ml**

92. Require ..... water to excret 1g of urea_____**0.05L**

93. Salivary and pancreatic secretion contain_____ **HCO3 ions**

94. Goblet produce in_____ **intestine**.

95. .....largest water soluble nutrient_____**VitaminB12**

96. True about action potential____ inflow of sodium ion_____ (**depolarization**)

97. Not a part of organ corti_____ **Reissner's men**

98. Cold receptor are.....more than heat receptor_____**3.5 times**

99. Human retina has ....cone_____**6million** 100.

Diameter pitutary gland_____**1cm**

101. . …….. are stimulated by stretch of the arterial wall._____**Baroreceptors**

102. Which is nitrogenous base..... urea..uric acid.. amonia_____**all of these**

103. ……. is the gastric proteolitic enzyme_____**pepsin**

104. gastric secretìn in mammals occurs in..... phases_____**3**

105. the temperature of …….. heterotherms changea on daily basis or annual basis._____ **temporal**

106. ……..provide heat insulation to mammals._____**fur**

107. . ……. involves use of respiratory system to lose heat by evaporating cooling_____**panting**

108. …….. are types of instinctive behaviour._____ **fixed action paterens**.

109. Sensory sensillia is a organ of ------ in insect_____**Taste**

110. In hair cell 20-300 are_____**nonmotilestereocilia**.

111. Camel lose---------water in body fluids_____**40%**

112. Mineralocorticoids include_____ **effects on mineral metabolism.**

113. An adaptive change happens due to chronic exposure to nex,naturally occuring environment known _____ **Acclimatization**.

114. Respiratory epithelium have only a thickness of_____**0.5 to 15um.**

BEST OF LUCK